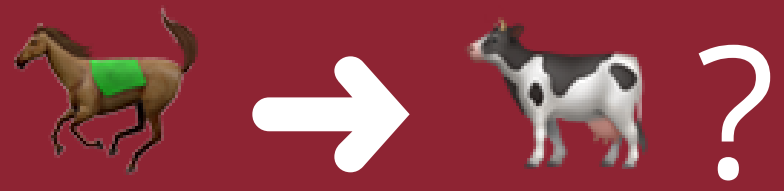
IS511E – Blockchain in Business
Midterm 1 – Group project

MeatTrace

How a shared ledger could stop meat fraud in the food supply chain

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Rabat Business School – 22 October 2026



Horse or beef?

In 2013, nobody could say where the label changed.

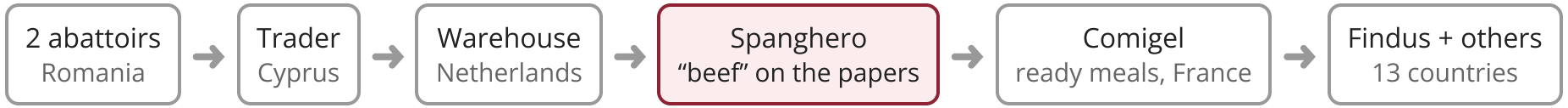
The problem: horse meat sold as beef

- In 2013, around **750 tonnes** of horse meat were sold as beef in Europe*
- It ended up in about **4.5 million products** (Findus lasagne…), from 28 companies in 13 countries*
- EU tests: **4.66 %** of beef products had horse DNA (193 out of 4,144)**

And today?

Food fraud is still here: in a 2022 EU operation, **46 %** of imported honey samples were adulterated, some with fake traceability documents***

How the meat travelled:



The abattoirs say the meat was labelled “horse”, French investigators blamed Spanghero (which denied it)*. **We chose this case because every company blamed the next one, and nobody could quickly prove where the label changed.**

* AFP via IOL, “Horsemeat: French company blamed”, 14/02/2013 and Al Jazeera, 15/02/2013 ** European Commission, horse meat test results 2013 *** OLAF (EU anti-fraud office), report 2023, “No money for fake honey”

Who is involved, and what goes wrong today

Stakeholder	What they want
Abattoir	Sell batches, prove it labelled correctly
Trader	Buy and resell fast, keep prices secret
FreshMeals (our client)	Avoid recalls and bad press
Retailers	Proof of origin for their customers
DNA lab	Be trusted by everybody
Food safety authority	Fast audits during a crisis

Today

- Certificates on paper or PDF, sent by email
- Each company keeps its own system (ERP)
- To trace a batch: phone calls and emails, one company at a time. It takes days (Walmart needed almost 7 days for mangoes*)
- The law only asks each firm to know its supplier and its customer ("one step back, one step forward")**, so nobody sees the whole chain

Our goal: trace any beef batch back to its abattoir in **less than 1 hour**, with every handover signed.

* LF Decentralized Trust (Hyperledger), Walmart case study ** EU Regulation 178/2002, article 18

Our solution: MeatTrace

To make the case concrete, we imagined a client: **FreshMeals**, a French ready-meal company (fictional) with €120M sales, 6 beef suppliers in 3 countries and about 1,500 batches a year (our assumptions).

How it works for each batch

1. The sender records the batch and **signs** it → **2.** The receiver checks it and **confirms** → **3.** A lab **tests** some samples (DNA) → **4.** Everyone allowed can **trace** the batch in minutes

What it brings

- **Faster:** minutes instead of days
- **Safer:** a relabelled batch is stopped at the next step
- **Fewer disputes:** one shared history
- **Compliance** with EU traceability rules

Why would partners join?

- **Retailers:** proof of origin for customers
- **Honest suppliers:** it's free, and it proves they were honest
- **FreshMeals:** smaller recalls, faster audits
- **The lab:** more tests to do

Users: abattoirs, traders, FreshMeals quality team, retailers, the DNA lab, the food authority (read only).

Business model and KPIs

Who pays?

- We chose a **consortium model**: 4 founding members (FreshMeals, 2 retailers, the DNA lab) share all the costs, about **€25k each per year** (our estimate)
- Suppliers join for free. We think getting them on board matters more than fees
- Later, new buyers pay a membership fee

Rollout

Year 1: pilot with FreshMeals, 3 suppliers and the lab
Year 2: all 6 suppliers + 2 retailers
Year 3: other buyers of the same suppliers

Resources: IT partner, hosting of 4 nodes, security, training, support, a governance board.

Our 3 KPIs

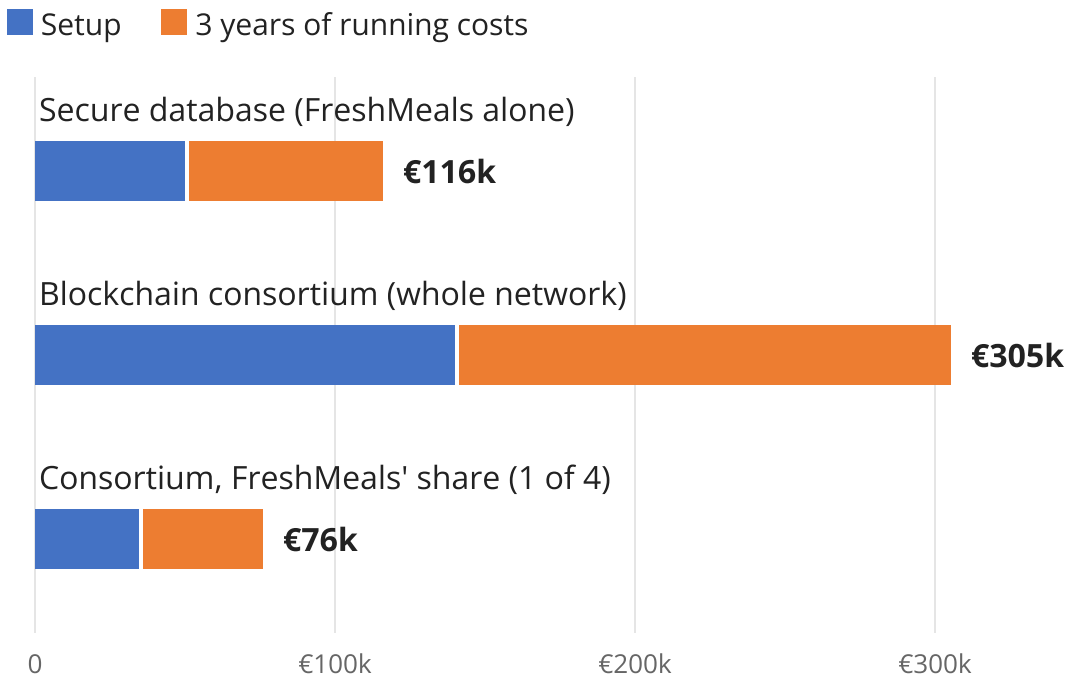
KPI	Today	Target
Time to trace a batch to the abattoir	Days*	**< 1 hour**
Batches with every handover signed	0 %	**100 %**
Batches with a linked DNA test	Rare	**≥ 10 %**

Expected benefits

- A recall only hits the guilty batches, not the whole product line (a full recall ≈ €2M, our assumption)
- Less staff time spent on tracing
- Carrefour said its blockchain-tracked chicken sold better (no figures given)**

* Order of magnitude from the Walmart case study (LF Decentralized Trust) ** ESM Magazine, “Carrefour says blockchain tracking boosting sales of some products”, 2019

3-year costs: database vs blockchain



	Setup	Per year	3 years
Database	€50k	€22k	**€116k**
Consortium	€140k	€55k	**€305k**
1 share of 4	€35k	€13.75k	**€76k**

How we estimated: €500 per developer day (≈80 days for the database, ≈180 for the blockchain), hosting, security audit, training. No crypto fees (private network). Details in the appendix.

What we conclude: blockchain costs €189k more in total, but FreshMeals pays less if 4 members share. If FreshMeals had to pay alone, we would choose the database.

All figures are our own estimates in euros. 3-year cost = setup + 3 × yearly cost.

Why we chose a permissioned consortium

Option	Good points	Problem for our case	Our choice
Public blockchain	Anyone can verify	Everyone sees our batches and partners; fees; slower	❌
Private blockchain run by FreshMeals	Fast, private	FreshMeals would be “judge and party”	❌
Permissioned consortium	Every member has a copy, nobody can rewrite alone, rights per role, a few seconds per block	More coordination; €305k but shared by 4; needs a board	✅ **chosen**
Secure database run by FreshMeals	Cheapest (€116k), fastest, simple	Partners must trust FreshMeals' admins, who could edit the history	Plan B

When a database is enough

When one trusted company runs the whole process. Then it is cheaper and simpler, and we would choose it.

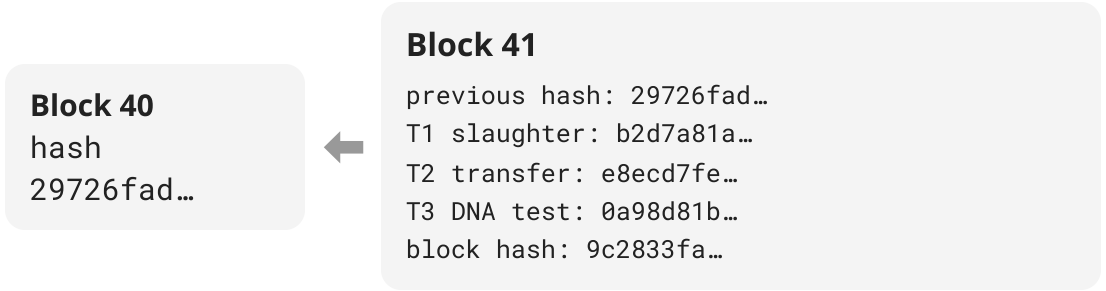
Why not here

6+ companies in 3+ countries, some of them competitors, blamed each other in 2013. Nobody would accept that a competitor controls the data.

Transactions, blocks, hash and signature

The 3 transactions we record

- **Slaughter:** batch_id, species, weight_kg, site, date, signed by the abattoir
- **Transfer:** batch_id, from, to, species_declared, weight_kg, signed by the sender, confirmed by the receiver
- **DNA test:** batch_id, lab, result, hash of the lab report, signed by the lab
- **Hash (SHA-256):** a 64-character fingerprint of these fields. One letter changes, the fingerprint changes
- **Signature:** the sender signs with its private key, the others check with its public key. It proves *who* sent it and that it was *not changed*
- **Off-chain:** prices, contracts and full lab reports stay with each company, only their hash is shared



Real hashes from our Python code.

Workflow: create ➜ check the role ➜ verify hash + signature ➜ validators agree ➜ shared history for audits

⚠️ A signature proves **who** said something, not that it is **true**. If a partner signs something false, random DNA tests catch it, and the contract says who is responsible.

Who can do what? Access rights and governance

Role	Create	Approve	Read	Validate blocks
Abattoir	Slaughter records	❌	Own batches	❌
Trader	Transfers	❌	Batches it held	❌
FreshMeals quality manager	Receptions	✅ not its own	Batches it buys	✅
DNA lab	Test results	✅ what it tested	Batches it tests	✅
Retailer	❌	❌	Origin of what it sells	✅
Food authority	❌	❌	Everything (audits)	Observer

Consensus: 3 out of 4

The 4 validators (FreshMeals, 2 retailers, the lab) take turns proposing blocks. A block is accepted when 3 of 4 sign it (like PBFT), so one dishonest member can't fake or block it. We know all the members, so no mining (PoW) or staking (PoS).

Our rules

- **Identity:** company certificate + personal login with 2FA
- **Nobody approves their own record**
- **Confidential:** prices and contracts stay off-chain
- **Revocation:** people who leave are removed the same day
- **Board:** the 4 founders, 1 vote each; new members or new rules need 3 votes; DNA results settle disputes

Python demo 1: hashing (integrity)

```
[4]  relabelled_record = {**slaughter_record, "species": "BEEF"}   # new dict
     relabelled_hash = record_hash(relabelled_record)

     print("Approved hash  :", approved_hash)
     print("Relabelled hash:", relabelled_hash)
     print("Same record?   :", approved_hash == relabelled_hash)
```

```
Approved hash  : b2d7a81a41a3ae8e3847b21c821538803f7268b85e502cb86736d78f6d2272e6
Relabelled hash: b2c60303ead9a020cc82bbf4de7292c4885478ab67f2ec54c967d8992228bc52
Same record?   : False
```

```
[7]  # the 2013 case: a NEW transfer document says "BEEF"
     signed_species = slaughter_record["species"]
     status = "OK" if resale_transfer["species_declared"] == signed_species \
              else "MISMATCH, batch blocked"
     print("Check                 :", status)
```

```
Check                 : MISMATCH, batch blocked
```

What we see

- We only changed one word (HORSE ➜ BEEF) and the hash is completely different
- Every validator keeps the original hash, so nobody can secretly edit a signed record
- In 2013 the fraud was a *new* "beef" document, so we added a rule: the declared species must match the signed history
- **Limit:** it proves nothing was changed, not that the abattoir told the truth

Screenshot of our notebook (IS511E_Midterm1_MeatTrace.ipynb), cells 4 and 7. Cell 7 is shortened here.

Python demo 2: who is allowed?

```
[10]  def can_do(user, action, record_id=None):
          account = users[user]
          if not account["active"]:
              return False, "account revoked"
          if action not in role_permissions[account["role"]]:
              return False, f"role '{account['role']}' has no '{action}' right"
          if action == "approve_record" and records[record_id]["created_by"] == user:
              return False, "separation of duties: cannot approve own record"
          return True, "permission granted"
```

```
[11]  tests = [("anna", "approve_record", "REC-001"), ("claire", "read_batch", "REC-001"), ...]
      ...
      display(pd.DataFrame(results))
```

	Result	User	Role	Action	Record
0	ALLOWED	anna	lab_inspector	approve_record	REC-001
1	ALLOWED	claire	quality_manager	read_batch	REC-001
2	DENIED	pieter	trader	edit_record	REC-001
3	DENIED	claire	quality_manager	approve_record	REC-007
4	DENIED	dan	trader	create_record	-

Why the 3 refusals matter

- ❌ **Pieter** (trader) can't edit the slaughter record: no relabelling like in 2013
- ❌ **Claire** can't approve her own reception: no self-approval
- ❌ **Dan** left his company: his account is revoked, he can't sign for them

✅ Anna (lab) approves the abattoir's record and Claire reads the batch history: normal work.

6 users, 5 roles. In real life this check would run on every node, not in a notebook.

Screenshot of our notebook, cells 10 and 11 (cell 11 shortened, “Reason” column hidden).

Roadmap, risks and our recommendation

Oct 2026
Proof of concept (done): our Python tests

→

Jan–Jun 2027: pilot
FreshMeals, 3 suppliers, the lab, one product (beef lasagne). ≈ €70k

→

2027–2028: deployment
6 suppliers, 2 retailers, authority as observer. ≈ €70k + €55k/year

→

2028+: improve
KPIs every quarter, add chicken and pork

Main risks

- **False data entered:** random DNA tests + the receiver must confirm
- **Data leaks:** rights per role, prices stay off-chain
- **Members disagree:** written rules, 3-of-4 vote
- **Costs too high:** pilot first, database as plan B
- **Partners don't join:** free for suppliers, simple website. TradeLens (300+ members) announced its closure in 2022 because the industry didn't join*

Our recommendation

Build MeatTrace as a permissioned consortium (FreshMeals, 2 retailers, a DNA lab): **€305k over 3 years, €76k for FreshMeals**.

First pilot goal: 100 % of pilot batches traced to the abattoir in under 1 hour.

Limits: it can't see inside the meat (we still need DNA tests), and without 3 co-funders we would go for the database. Still to build: real identities, nodes, website, security audit.

* Container Magazine, “Maersk and IBM to discontinue TradeLens following lack of industry collaboration”, 30/11/2022. Costs per stage are our estimates.

Sources

- AFP via IOL (14/02/2013), *Horsemeat: French company blamed*, iol.co.za/news/world/horsemeat-french-company-blamed-1470925
- Al Jazeera (15/02/2013), *France says horse meat tip-offs ignored*, aljazeera.com/news/2013/2/15/france-says-horse-meat-tip-offs-ignored
- European Commission, *Horse meat 2013–14: test results*, food.ec.europa.eu
- OLAF (2023 report), *No money for fake honey*, ec.europa.eu/olaf-report/2023
- EU Regulation 178/2002, article 18 (traceability), eur-lex.europa.eu
- LF Decentralized Trust, *Walmart case study* (Hyperledger Fabric), lfdecentralizedtrust.org/case-studies/walmart-case-study
- Container Magazine (30/11/2022), *Maersk and IBM to discontinue TradeLens*, container-mag.com
- ESM Magazine (2019), *Carrefour says blockchain tracking boosting sales of some products*, esmmagazine.com
- IS511E course, sessions 1 to 5

FreshMeals, its users and all business-plan numbers are our own assumptions.

Appendix: how we estimated the costs

Secure database

Item	Setup	Per year
Website + database (≈80 days × €500)	40,000	
Security (2FA, encryption, test)	6,000	
Supplier training	4,000	
Cloud hosting + backups		6,000
Maintenance (≈24 days)		12,000
Security monitoring + audit		4,000
Total	**50,000**	**22,000**

3 years = 50,000 + 3 × 22,000 = **€116,000**

Permissioned consortium (4 nodes)

Item	Setup	Per year
Ledger + smart contracts + website (≈180 days)	90,000	
Network, identities, certificates	20,000	
External security audit	15,000	
Legal charter + onboarding	15,000	
Hosting (4 nodes × €5,000)		20,000
Maintenance (≈50 days)		25,000
Board, audits, certificates		10,000
Total	**140,000**	**55,000**

3 years = 140,000 + 3 × 55,000 = **€305,000** (1 share of 4 = €76,250)